WORKED SOLUTION

STEP I 2008, Question 8

Pure • Differential equation

> **STEP I 2008, Question 8**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/08-s1-q8

The key idea

Both equations involve y' but not y'', and they are not the kind we can separate or integrate directly. Differentiating once more is the trick: the terms in y' alone cancel, and what is left factorises with y'' as one factor. Each factor gives a case, and in each case we substitute back into the *original* equation rather than integrating, which pins down the arbitrary constant for free.

Solution

Throughout we write $p = y'$, so $p' = y''$.

Part (i)

Differentiate $p^2 - xp + y = 0$ with respect to x, using the product rule on xp:

$$2p\,y'' - (p + x\,y'') + p = 0 \quad \Longrightarrow \quad y''\,(2p - x) = 0.$$

The two p terms cancel, so at each point either $y'' = 0$ or $2y' = x$, as required.

Case $y'' = 0$. Then y' is a constant m and $y = mx + c$. Substituting into $(*)$:

$$m^2 - xm + (mx + c) = 0 \quad \Longrightarrow \quad c = -m^2.$$

So the curve is the straight line $y = mx - m^2$.

Case $2y' = x$. Put $p = \frac{1}{2}x$ into $(*)$:

$$\tfrac{1}{4}x^2 - \tfrac{1}{2}x^2 + y = 0 \quad \Longrightarrow \quad 4y = x^2.$$

> **Ask yourself** Why not just integrate $y' = \frac{1}{2}x$ to get $y = \frac{1}{4}x^2 + k$?

We could, but differentiating $(*)$ threw information away: any $y = \frac{1}{4}x^2 + k$ satisfies $2y' = x$, yet only $k = 0$ satisfies $(*)$ itself (for example, $k = 1$ at $x = 0$ gives $0 - 0 + 1 \neq 0$).

Substituting straight back into $(*)$ uses all the information at once and gives the constant directly. The same remark explains why $c = -m^2$ in the first case.

So the curve is either a line $y = mx + c$ with $c = -m^2$, or the parabola $4y = x^2$.

The two answers are linked: the line $y = mx - m^2$ meets $4y = x^2$ where $x^2 - 4mx + 4m^2 = 0$, that is $(x - 2m)^2 = 0$, so each line is the tangent to the parabola at $(2m, m^2)$. The parabola is the *envelope* of the family of lines.

Part (ii)

Differentiate $(x^2 - 1)p^2 - 2xyp + y^2 - 1 = 0$ with respect to x:

$$2xp^2 + 2(x^2 - 1)p\,y'' - 2yp - 2xp^2 - 2xy\,y'' + 2yp = 0.$$

The terms without y'' cancel in pairs, leaving

$$2y''[(x^2 - 1)p - xy] = 0.$$

Case $y'' = 0$. Then $y = mx + c$. It helps to notice that the equation can be rearranged as

$$(y - xp)^2 = 1 + p^2,$$

since $(y - xp)^2 - p^2 - 1 = y^2 - 2xyp + (x^2 - 1)p^2 - 1$. For the line, $y - xp = c$ and $p = m$, so

$$c^2 = 1 + m^2, \qquad \text{that is,} \qquad y = mx \pm \sqrt{1 + m^2}.$$

Case $(x^2 - 1)p = xy$. Where $x^2 \neq 1$ we have $p = \dfrac{xy}{x^2 - 1}$, so

$$y - xp = y - \frac{x^2 y}{x^2 - 1} = \frac{-y}{x^2 - 1}.$$

Substituting into $(y - xp)^2 = 1 + p^2$ and multiplying by $(x^2 - 1)^2$:

$$\begin{aligned} y^2 &= (x^2 - 1)^2 + x^2 y^2 \\ y^2(1 - x^2) &= (1 - x^2)^2 \\ y^2 &= 1 - x^2, \end{aligned}$$

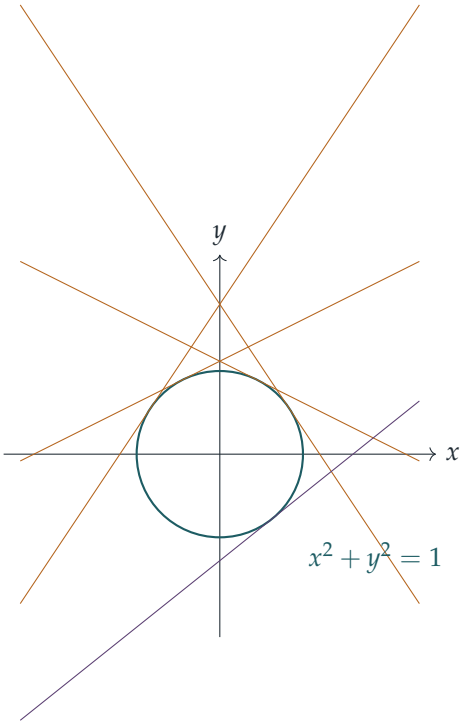
dividing by $1 - x^2 \neq 0$. So the curve is the circle $x^2 + y^2 = 1$. (At $x = \pm 1$ the condition $(x^2 - 1)p = xy$ forces $y = 0$, giving the points $(\pm 1, 0)$, which lie on the same circle.)

Conversely the circle does work: on it $p = -x/y$, so $y - xp = (y^2 + x^2)/y = 1/y$ and $1 + p^2 = (x^2 + y^2)/y^2 = 1/y^2$, which agree.

So the curve is either a straight line $y = mx + c$ with $c^2 = 1 + m^2$, or the unit circle $x^2 + y^2 = 1$.

Ask yourself What do the lines $y = mx \pm \sqrt{1 + m^2}$ have to do with the circle?

The distance from the origin to $y = mx + c$ is $|c|/\sqrt{1 + m^2}$, which here is exactly 1. So the lines are precisely the (non-vertical) tangents to the unit circle, just as the lines in (i) were the tangents to the parabola.



Checks

- Take $m = 1$ in (i): $y = x - 1$ has $p = 1$, and $1 - x + x - 1 = 0$. It touches $4y = x^2$ at $(2, 1)$.
- In (ii), the point $(\frac{3}{5}, \frac{4}{5})$ on the circle has $p = -\frac{3}{4}$; then $(x^2 - 1)p^2 - 2xyp + y^2 - 1 = -\frac{16}{25} \cdot \frac{9}{16} + \frac{18}{25} + \frac{16}{25} - 1 = 0$.
- Numerically, we checked both derivative factorisations along random cubic curves, and confirmed the lines, parabola and circle give residual 0 (to 10^{-13}), while $y = \frac{1}{4}x^2 + 1$ and $x^2 + y^2 = 4$ do not.

Common slips

- Integrating $y' = \frac{1}{2}x$ or $y'' = 0$ and leaving an arbitrary constant: the constant must be fixed by the original equation.
- Losing the $\pm$ in $c = \pm\sqrt{1 + m^2}$: there are two parallel tangents of each gradient.

Beyond STEP

Part (i) is a *Clairaut equation*, $y = xp - f(p)$: its general solution is the family of lines $y = mx - f(m)$, and it also has a *singular solution*, the envelope of that family, which

no choice of constant produces. Part (ii) is the same phenomenon, $y = xp \pm \sqrt{1+p^2}$, with the circle as envelope. Singular solutions show that a first-order equation need not have a unique solution through each point; when uniqueness holds is the subject of the existence and uniqueness theorems met in a first ODE course. Boyce and DiPrima, *Elementary Differential Equations*, is a good place to read about this.

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